Geometric Dimensioning & Tolerancing (GD&T)

Geometric Dimensioning and Tolerancing, or GD&T, is used to overcome the limitations of traditional dimensioning, by controlling the shape of the different parts and how close they are to an ideal geometric form.

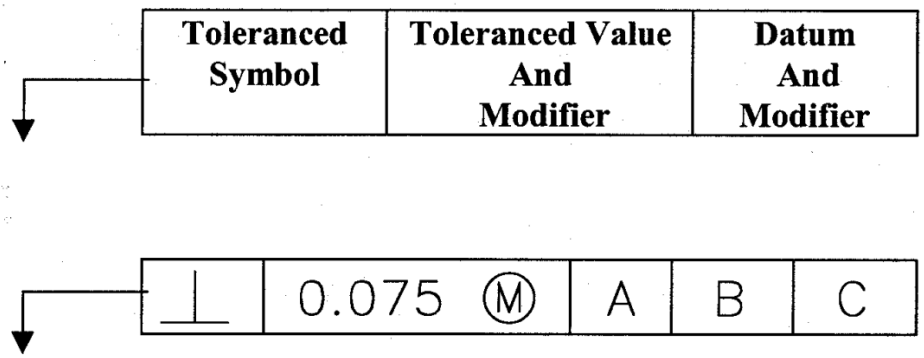
Shown are the symbols used for GD&T:

Table 1.27 Symbols for geometric characteristics

Geometric characteristic	*Symbol*	*Type of tolerance*	*Application*
Straightness	—	Form	For individual features
Flatness	▱		
Circularity	○		
Cylindricity	⌭		
Profile of a line	⌒	Profile	For individual or related features
Profile of a surface	⌓		
Angularity	∠	Orientation	For related features
Perpendicularity	⊥		
Parallelism	//		
Position including concentricity and symmetry	⌖	Location	
Circular runout	↗	Runout	
Total runout	⌰		

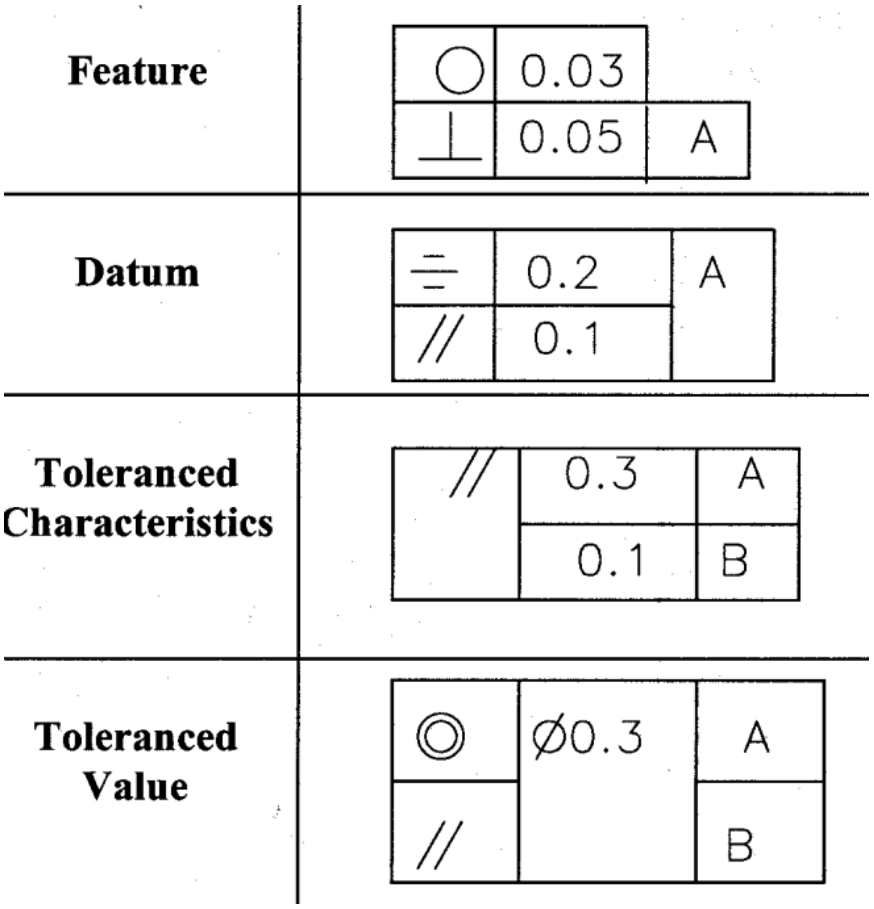
Feature Control Frame

To apply geometric tolerances, we use a feature control frame, with 2 to 5 rectangle segments:



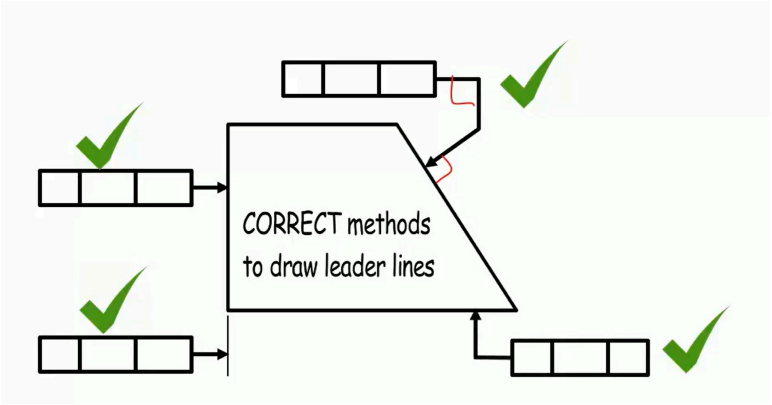
If the tolerance is linked to a dimension, it is placed on top of the FCF.

FCFs can also be stacked and merged, if any cells match:



For leader lines, they must be on the left or right of the FCF, normal to the outline of the drawing and the FCF, but multiple bends may be used.

leader lines may be placed on extension lines too.

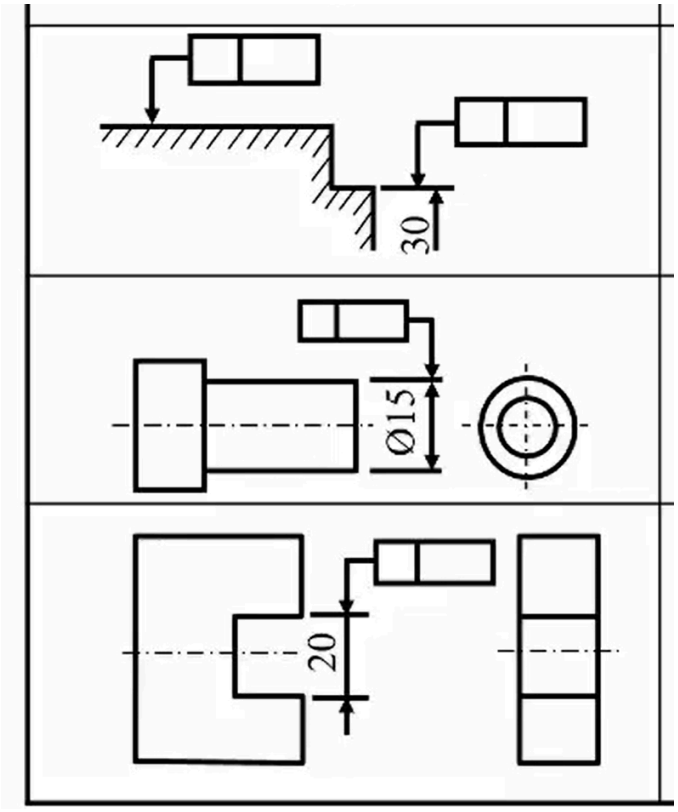


Where the FCF is placed refers to what geometry it is controlling.

When the leader line is on a flat surface (or the extension line of one), the FCF controls that surface.

When the leader line points to the tip of a circular dimension feature, it controls the cylindrical axis of the cylinder.

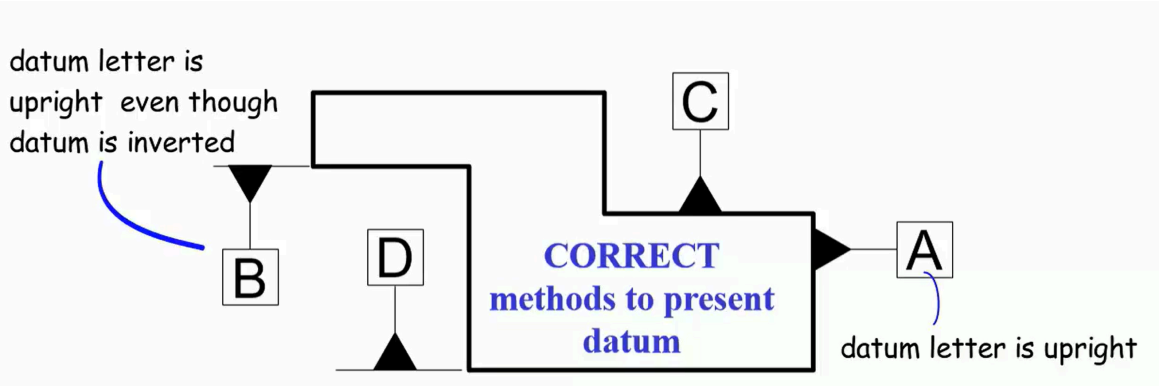
When the leader line points to the tip of a non circular dimension feature, the midplane is controlled.



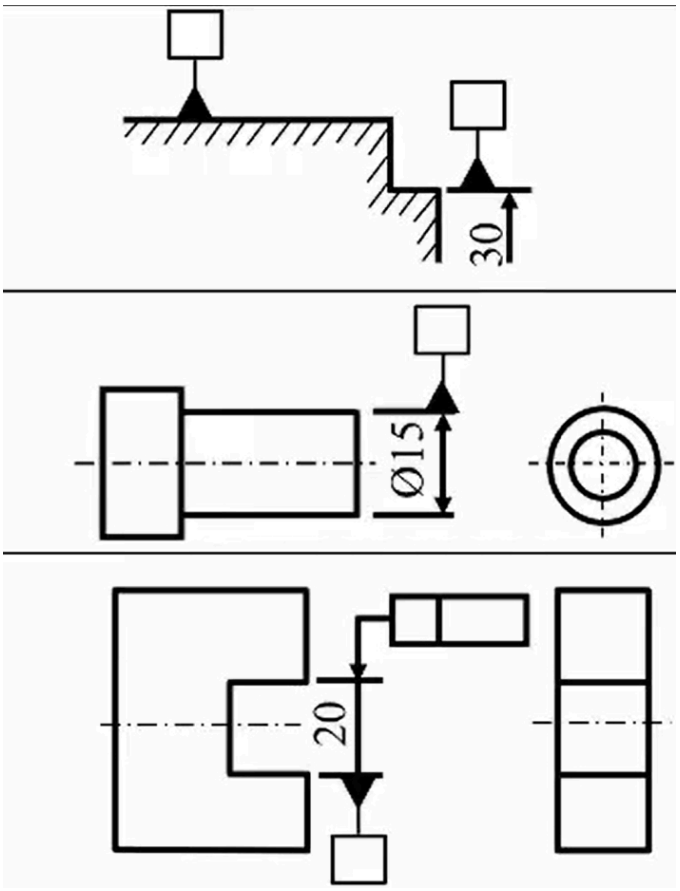
Datums

Datums are reference planes for some geometric tolerances. they are drawn with a dark triangle with the flat end flush with the outline or an extension

line, and connected to a box with an upright letter.

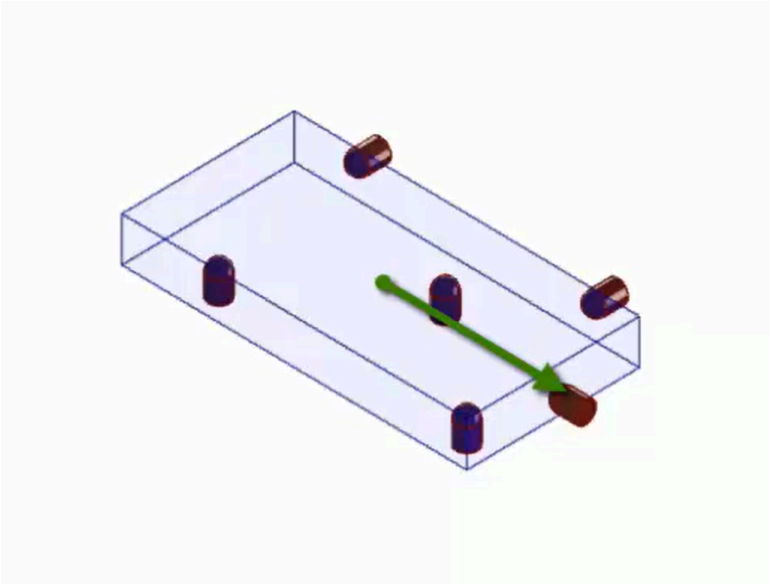


Like FCFs, datums can be defined similarly to FCFs:



Datums are indicated in FCFs at the end, in order of their importance from primary to tertiary.

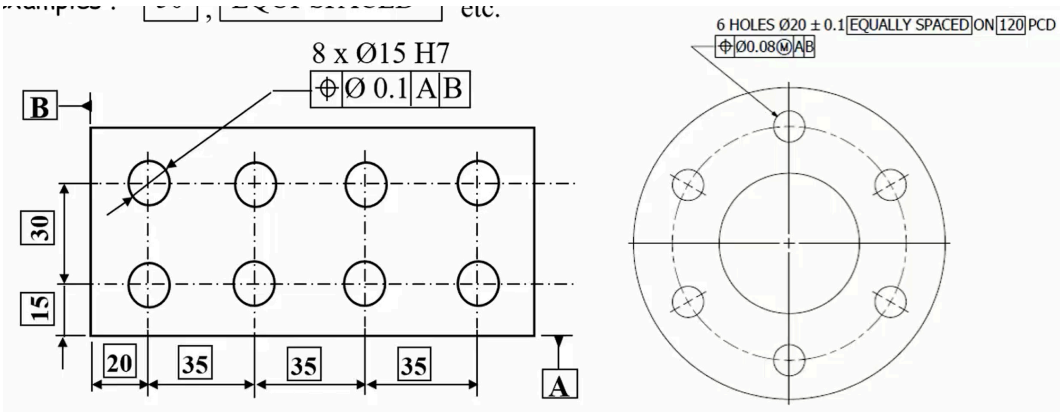
The way datums are prioritised is by locating the part in order of the datums chosen.



the first datum is set with 3 points to define a plane, then 2 points to define the second datum, and a single point for the last plane.

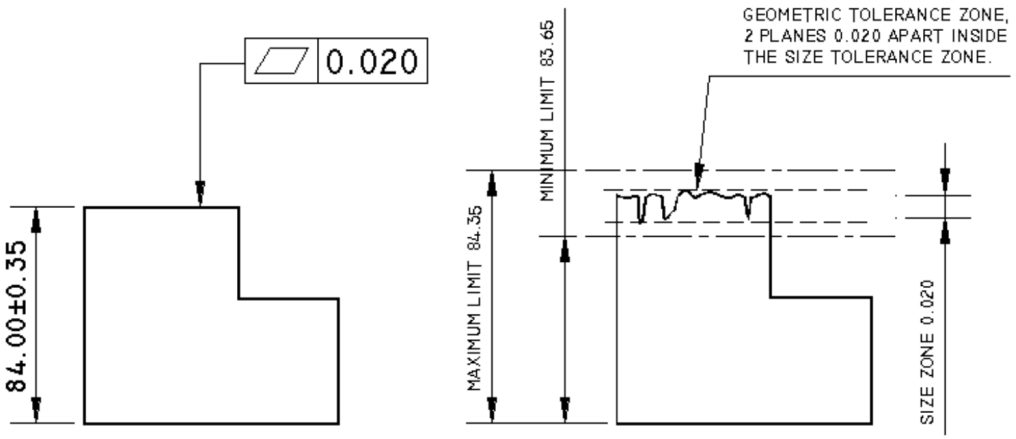
Basic Dimensions/Theoretical Exact Dimensions

In the case when position, profile and angularity is toleranced, the nominal dimension is no longer the toleranced parameter, hence we box it to show it represents the basic dimension.



Combined Tolerances

In the case where both dimensional and geometric tolerances are applied, the geometric tolerance is typically finer than the dimensional one.



GD&T Interpretations (of tolerance x)

The first 4 tolerance types do not need any datums, and other tolerance types may have datums applied.

Straightness

If applied to a surface, any line on that surface must lie between two parallel lines x apart

For cylindrical axes, it must lie in cylinder of diameter x

for midplanes, it must lie between two parallel planes x apart

Flatness

The surface/midplane must lie between two parallel planes x apart

Circularity

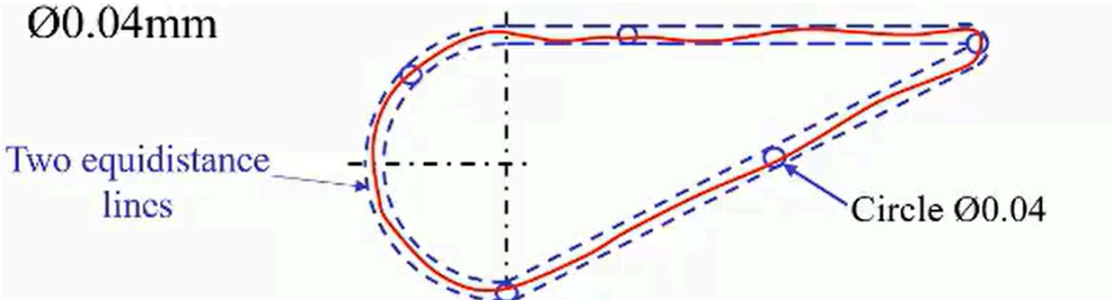
The circular profile must lie between two concentric circles of radius gap x apart

Cylindricity

The cylindrical surface shell must lie between two coaxial cylinders of radius gap x apart

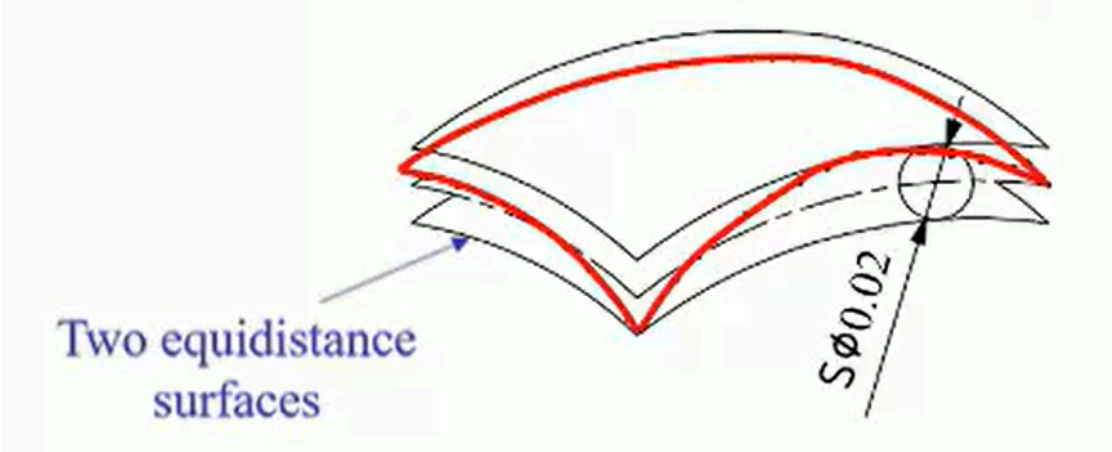
Profile

The 2D outline of any section must line between enveloping circles of diameter x



Surface Profile

Similar to profile, except the whole surface lies between enveloping spheres of diameter x



Parallelism/Perpendicularity/Angularity

Like flatness, but the planes are parallel/perpendicular/at the true angle to the specified datum

Position

Typically for holes, the axis lies between in a cylindrical zone of diameter x

Circular Runout

Similar to circularity, however the circles used must be centered on a datum, and checks for each section

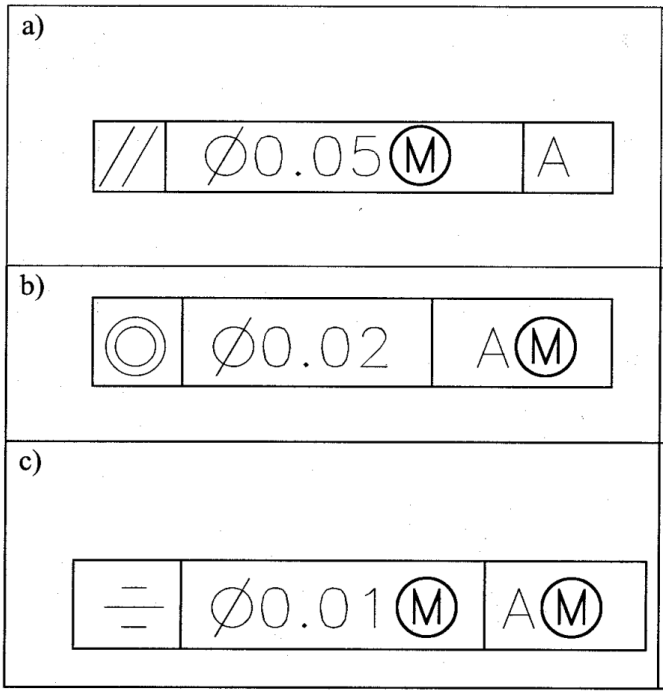
Surface Runout

similar to cylindricity but with a datum (?????)

Maximum Material Condition

Maximum Material Condition, or MMC, is a modifier applied to FCFs to allow for more tolerance when the part is smaller than the MMC.

MMC can be applied to either the tolerance, the datum, or both of an FCF.

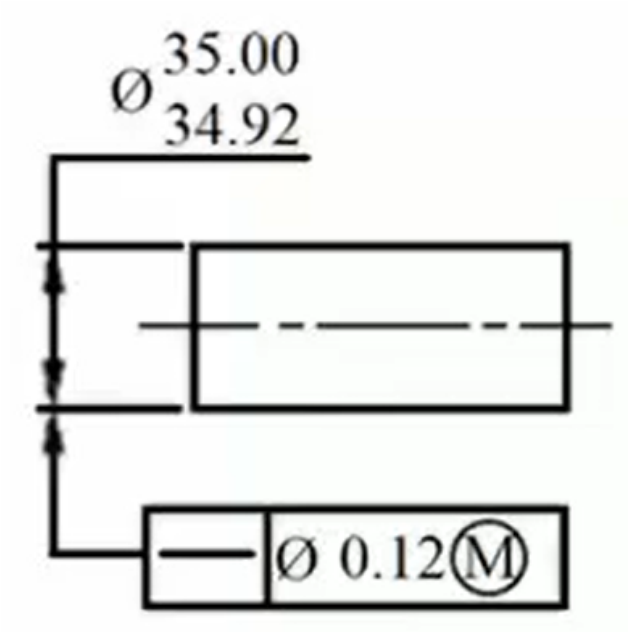


Firstly, the MMC dimension is based on the limits of the part itself. For shafts, the MMC dimension is the maximum limit while for holes the MMC dimension is the minimum limit.

Then, the absolute deviation from the MMC size is added onto the geometric tolerance value, increasing the tolerance zone. In other words,

$$\text{Effective Tolerance} = |\text{MMC Size} - \text{Actual Size}| + \text{Stated Tolerance}$$

For example, for this shaft, say it is slightly undersized at dia 34.98mm.



The deviation from the MMC condition is

$$35.00 - 34.98 = 0.02mm$$

Hence the new tolerance zone is now

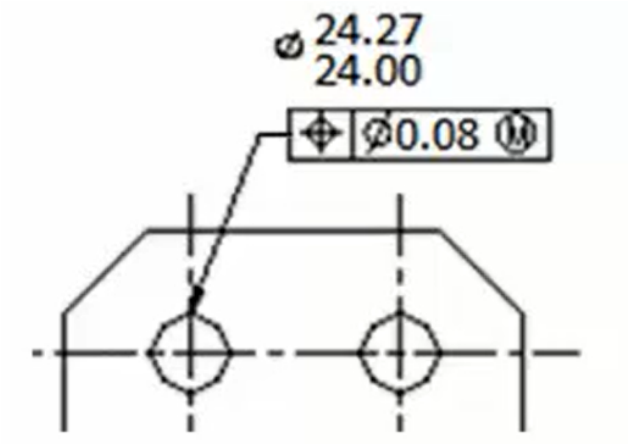
$$0.12 + 0.02 = 0.14mm \text{ diameter}$$

To check if the shaft also fits the tolerances, a ring gauge is also used, where the diameter is

$$\begin{aligned}\text{Ring Gauge} &= \text{MMC Size} + \text{MMC Tolerance} \\ &= \text{Actual Size} + \text{Effective Tolerance}\end{aligned}$$

which is always constant regardless of the deviation from the MMC. Hence any shaft within it's tolerance will pass through the ring gauge

A similar process works for holes:



Say the hole is oversized and has 24.05mm diameter.

The deviation is now

$$24.05 - 24.00 = 0.05mm$$

Hence the new tolerance zone is

$$0.08 + 0.05 = 0.13mm \text{ diameter}$$

To check if a hole is to tolerance, we use a Pin or Plug gauge, of diameter

$$\begin{aligned}\text{Pin Gauge} &= \text{MMC Size} - \text{MMC Tolerance} \\ &= \text{Actual Size} - \text{Effective Tolerance}\end{aligned}$$